

Restarts substitute for initialization in the Nelder–Mead method, and which restart trigger wins depends on dimension

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Abstract

Wessing (2019) showed that the initial simplex changes Nelder–Mead’s runtime by orders of magnitude, and that the effect survives Kelley’s oriented restarts. That robustness check held the restart scheme fixed, so the interaction was never quantified. We cross seven initialization procedures with six restart triggers and two restart constructions on the COCO/BBOB noiseless suite at $n \in \{2, 5, 10, 20\}$ (43 050 runs).

Restarts largely substitute for initialization: the spread in initializer mean rank falls from 2.68 under a single descent to 0.89–1.24 once a trigger is enabled, and the MATLAB `fminsearch` initializer—worst of seven on its own—becomes unremarkable. The substitution erodes with dimension.

Which trigger to use, however, depends strongly on n , and the two published detectors sit at opposite ends of that dependence. The simplex-degeneracy test of Luersen and Le Riche (2004) degrades monotonically, from 74.5% of problems solved at $n = 2$ to 4.2% at $n = 20$ against 11.4% for not restarting at all; a sweep of its tolerances over 40 orders of magnitude shows this to be a mis-setting rather than a property of the criterion, but also that no setting makes it better than a plain simplex-size test. The sufficient-decrease test of Kelley (1999) moves the other way, and is the only trigger whose advantage grows with dimension. Two trivial triggers—restart when the simplex is small, or when its vertex values are flat—are the most reliable across the range.

Keywords: Nelder–Mead · direct search · simplex initialization · restart strategies · derivative-free optimization · benchmarking

1 Introduction

The Nelder–Mead simplex method (Nelder and Mead, 1965) remains among the most widely used derivative-free optimizers despite the absence of general convergence guarantees (McKinnon, 1998; Lagarias et al., 1998). Two design choices govern its practical behaviour, and the literature has treated them separately.

The first is the initial simplex. Wessing (2019) showed that common initialization procedures inflate runtime by orders of magnitude even on a spherical objective, and recommended normalizing the search space and choosing the simplex independently of the starting point. Takenaga et al. (2023) studied initial simplex size and shape under a limited evaluation budget; Han and Neumann (2006) examined the interaction with dimension.

The second is the restart policy, and in particular its *trigger*: the test that decides the method has stopped making progress. Kelley (1999) detects stagnation by failure of a sufficient-decrease condition on the mean vertex value and remedies it with an oriented restart. Luersen and Le Riche (2004) restart on simplex smallness, on flatness of the vertex values, or on simplex *degeneracy*. Hansen (2009) benchmarked a two-level restart schedule and Pošík and Huyer (2012) compared restarted local searchers, Nelder–Mead among them.

These literatures do not meet. Wessing ran his comparison against Kelley’s restarting variant, but as a robustness check: the restart scheme was fixed and the interaction was not measured.

Nor, to our knowledge, has any trigger been ablated against the others. Both detectors above are motivated—one by a descent argument, one by simplex geometry—and neither has been compared to the trivial alternative of restarting when the simplex simply becomes small. We ask:

- (Q1) Do restarts substitute for initialization—and over what dimension?
- (Q2) Which trigger should fire the restart, and do the principled detectors earn their complexity?

Section 3 describes a factorial answering both. Because trigger and re-seeding are easy to confound—Kelley’s detector is published together with a deliberately *local* re-seed, while the others are naturally paired with a global one—we vary restart construction as a third factor rather than assuming it away.

2 Background: the two detectors

Let S^k have vertices ordered by objective value, let $D^k f$ denote the simplex gradient, and let

$$L(Y) = [y^1 - y^0, \dots, y^n - y^0] = [e^1, \dots, e^n]$$

be the edge matrix, with σ_+ and σ_- the largest and smallest edge length.

Kelley’s sufficient-decrease test. Kelley (1999) flags stagnation when the $k + 1$ st iteration fails

$$\bar{f}^{k+1} - \bar{f}^k < -\alpha \|D^k f\|^2, \quad (1)$$

where $\bar{f}$ is the *mean* of the vertex values—not the best, which a Nelder–Mead iteration is not guaranteed to improve—and $\alpha = \alpha_0 \sigma_+(S^0)/\|D^0 f\|$. A restart is performed when (1) fails but $\bar{f}^{k+1} - \bar{f}^k < 0$. The remedy is an *oriented restart*: a smaller simplex with orthogonal edges of length $\frac{1}{2}\sigma_-(S^k)$ aligned against the sign pattern of $D^k f$.

GBNM’s degeneracy test. The globalized bounded Nelder–Mead of Luersen and Le Riche (2004) declares a simplex degenerate—and reinitializes it—when it is neither small nor on a bound and

$$\frac{\min_k \|e^k\|}{\max_k \|e^k\|} < \varepsilon_{s3} \quad \text{or} \quad \frac{|\det L(Y)|}{\prod_k \|e^k\|} < \varepsilon_{s4}. \quad (2)$$

The second quantity is the Hadamard ratio, equal to 1 for orthogonal edges and 0 for linearly dependent ones, independently of length. Both are translation- and scale-invariant, which is what makes them usable as a shape diagnostic. Equation (2) is thus a conditioning-triggered restart, predating the geometry-driven model-improvement steps of model-based derivative-free optimization (Conn et al., 2009). Related ideas constrain the *operations* instead: Tseng (2000) and Nazareth and Tseng (2002) hold the simplex volume above a threshold so interior angles stay bounded away from zero.

A separate line restores convergence by constraining the simplex directly rather than restarting it: Price et al. (2002) poll a positive basis, Bűrmen et al. (2006) restrain the simplex to a grid, and Gao and Han (2012) adapt the coefficients to dimension—the second variant on which Wessing (2019) confirmed his initialization effect. Coefficient choice has been revisited since Barton and Ivey (1996), and Wang and Shoup (2011) give a parameter sensitivity study of the method—but of the reflection, expansion and contraction coefficients and the initial simplex size, not of any restart tolerance. None treat the trigger as the object of study.

Luersen and Le Riche report that ε_{s3} and ε_{s4} “may be difficult to tune, so that a simplex which is becoming small may be tagged as degenerated before,” and add a rule requiring two consecutive detections to compensate; the same machinery carries into their constrained variant (Luersen et al., 2004). That admission appears not to have been followed up.

3 Experimental design

Factor A: initialization (7 levels). `pfeffer`, the MATLAB `fminsearch` default and the principal subject of Wessing (2019); the regular simplex of Spendley et al. (1962), which is also GBNM’s Eq. (A.1), at edge scales of 100%, 10% and 2% of the smallest box dimension, the latter two spanning the range from which GBNM draws restart simplices; `uniform` and `gaussian` random simplices; and a minimal positive spanning basis $W = ZB^-$.

Factor B: restart trigger (6 levels). The six below are the triggers in current use. One classical alternative is not included: the factorial test of O’Neill (1971), which perturbs each coordinate at the converged point to check it is a local minimum and restarts if it is not. It costs $2n$ evaluations per invocation rather than none, which puts it in a different class from the six here, and we leave it to future work. `none` (single descent, terminated rather than restarted); `size` (small simplex); `flat` (small spread of vertex values); `conditioning` (Eq. (2)); `gbnm` (size $\vee$ conditioning, as published); and `progress` (Eq. (1)).

Factor C: restart construction (2 levels). `gaussian_best`, a fresh simplex drawn about the incumbent—a global move on a bounded box; and `oriented`, Kelley’s Eq. (8.7)—a deliberately local move. `none` takes no construction and `progress` appears only in its published pairing, giving $7 \times (1 + 1 + 4 \times 2) = 70$ arms.

Benchmark. COCO/BBOB noiseless (Hansen et al., 2021), f_1 – f_{24} , $n \in \{2, 5, 10, 20\}$, instances 1–5: 480 problems $\times$ 70 arms = 33 600 runs at a budget of $2000n$ evaluations. BBOB shifts and rotates each instance, so a structured simplex centred in the box gains no unearned advantage. This COCO build does not expose f_{opt} , so the reference is f_L , the best value reached by any arm, and the convergence test is that of Moré and Wild (2009), $f(x_0) - f(x) \geq (1 - \tau)(f(x_0) - f_L)$, at $\tau \in \{10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}\}$; this is the budget-aware alternative to the performance profiles of Dolan and Moré (2002). We additionally ran the same factorial on five unshifted analytic functions at $n \in \{2, 5, 10\}$ (15 seeds; 9450 runs), not as evidence but as a control on the benchmark itself. The control does not extend to $n = 20$, so the comparison in Section 4.8 is drawn over three dimensions rather than four.

Reporting rule. Results are reported per dimension. We pool across dimensions *only where the ordering is dimension-stable*, and say so at each point; the central finding below is that for several comparisons it is not, and a pooled number would then be an artifact of which dimensions happened to be run.

Statistical protocol. The unit of analysis is the *problem*, not the run: the seven initializers on one problem are not independent, so each problem is reduced to a single outcome (solved by a majority of the seven) before any test. Paired comparisons use an exact McNemar test on those outcomes, and Holm correction is applied across the entire family of 32 comparisons the paper reads, so no claim below rests on an uncorrected p -value. Effect sizes are reported as the paired win rate $P(\text{arm A ends lower} \mid \text{same problem})$, read on the conventional 0.56/0.64/0.71 small/medium/large scale.

Two choices deserve a word. We do not use a Wilcoxon signed-rank on the raw objectives: BBOB values carry per-instance offsets and span many orders of magnitude, so that test is dominated by whichever problems happen to have large $|f|$. And we do not use the unpaired Vargha–Delaney A_{12} , which across all cross-pairs compares one arm on one problem against another arm on a different problem; problem difficulty then swamps the arm effect, and it returns “negligible” for comparisons whose paired win rates run from 20% to 97%.

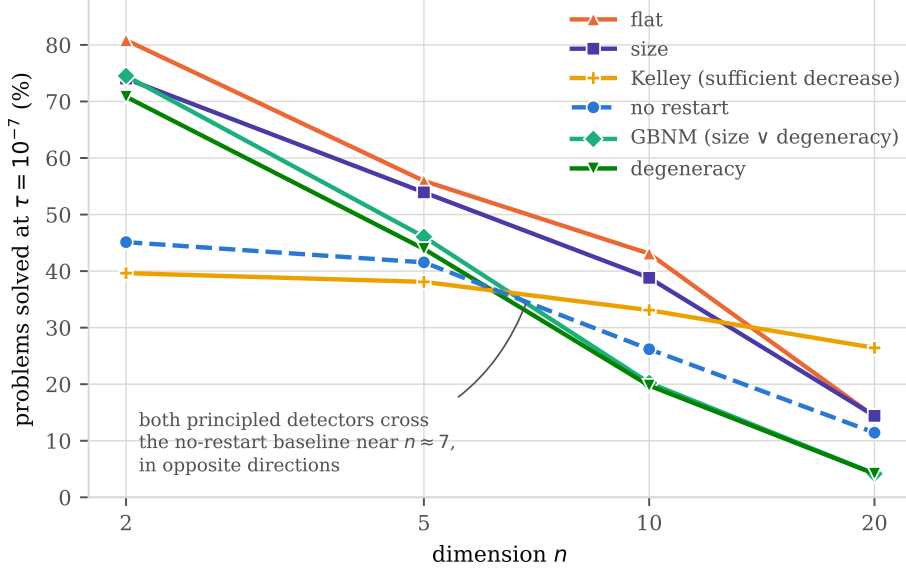


Figure 1: Problems solved at $\tau = 10^{-7}$ against dimension, at `gaussian_best` construction (`progress` in its published pairing). The two principled detectors cross the no-restart baseline near $n \approx 7$ in *opposite* directions.

Table 1: Problems solved at $\tau = 10^{-7}$ (%) and restarts per run, by dimension. Constructions are `gaussian_best` except `progress`, which appears in its published pairing with `oriented`.

trigger	solved at $\tau = 10^{-7}$ (%)				restarts per run			
	$n=2$	$n=5$	$n=10$	$n=20$	$n=2$	$n=5$	$n=10$	$n=20$
flat	80.8	56.0	43.1	14.3	53.8	40.9	27.3	11.8
size	74.0	53.9	38.8	14.4	33.7	28.0	18.0	7.3
gbnm	74.5	46.1	20.2	4.2	79.9	93.5	131.1	463.3
conditioning	70.8	43.9	19.8	4.2	64.0	86.7	126.4	462.5
progress	39.6	38.1	33.1	26.4	7.5	23.9	20.4	22.4
none	45.1	41.5	26.2	11.4	0	0	0	0

Finally, the paper’s dimensional claims are claims about a *trend*, and testing them as four separate point comparisons is both the wrong question and needlessly weak. We test them directly with a Cochran–Armitage trend test on the share of discordant pairs, with $\log_2 n$ as the dose.

Response. Besides solved fractions we log, per iteration, both quantities of Eq. (2) and which criterion fired at each restart. The Hadamard ratio is computed in log space via a signed log-determinant; evaluated directly it underflows to `NaN` at $n = 10$ once edge lengths reach 10^{-40} , that is, exactly in the degenerate regime of interest.

4 Results

4.1 The best trigger depends on the dimension

Figure 1 and Table 1 carry the paper’s main result: *the ranking of restart triggers is not stable across dimension*, and the two published detectors sit at opposite ends of that instability.

The degeneracy test degrades monotonically, from 74.5% at $n = 2$ to 4.2% at $n = 20$. It

starts above the no-restart baseline and ends far below it: at $n = 10$ it solves 20.2% against the baseline’s 26.2%, and at $n = 20$ it solves 4.2% against 11.4%—roughly a third as many problems as never restarting. Kelley’s detector moves the other way, from 5.5 points below the baseline at $n = 2$ to 15.0 points above it at $n = 20$. It is the only trigger whose advantage over not restarting grows with dimension, and the two cross the baseline in opposite directions at nearly the same place, $n \approx 7$.

Both trends are significant when tested as trends. The share of discordant problems favouring Kelley over no-restart rises $0.00 \rightarrow 0.18 \rightarrow 0.54 \rightarrow 0.77$ across the four dimensions (Cochran–Armitage $z = +4.44$, $p = 9.1 \times 10^{-6}$), while for the degeneracy trigger it falls $1.00 \rightarrow 0.55 \rightarrow 0.06 \rightarrow 0.00$ ($z = -7.80$, $p = 6.0 \times 10^{-15}$). Read instead as point comparisons, the picture is more uneven and we report it as such: the degeneracy trigger is significantly worse than no-restart at $n = 10$ and $n = 20$ after Holm correction ($p = 9.3 \times 10^{-3}$ and 3.5×10^{-4} ; at $n = 20$ it solves *no* problem that no-restart misses, against 17 the other way), whereas Kelley’s advantage at $n = 20$ does *not* survive correction as an isolated comparison ($p = 0.094$). The Kelley result is therefore a trend claim, and we make it only as one.

The two trivial triggers are the most reliable across the range, and both beat both principled detectors at every dimension from $n = 5$ upward. We do not rank them against each other: **flat** leads on the relative measure at $n \leq 10$ and **size** on the absolute one at $n = 10$, and a paired test separates them only at $n = 2$ (in **flat**’s favour, $p = 1.6 \times 10^{-3}$); everywhere else they are indistinguishable. At $n = 2$ **gbnm** and **size** are indistinguishable (74.5 vs. 74.0), so the advantage of the trivial triggers is a claim about $n \geq 5$, not about the whole range. Since that ordering *is* dimension-stable over $n \geq 5$, we state it pooled: adding Eq. (2) to a size test makes the combination worse, which is stronger than the test merely failing to help. That comparison survives correction at every dimension from $n = 5$ ($p \leq 2.7 \times 10^{-2}$), with paired win rates of 0.650, 0.633 and 0.692—a medium effect. The corresponding comparison against no-restart is large at every dimension (0.771 to 0.979).

Varying construction does not change these conclusions. Under **oriented** re-seeding the trivial triggers still lead at $n \leq 10$ (47.4–50.1% against 31.9–48.5%), and construction has a large main effect of its own—the local re-seed costs every trigger 11.5 to 16.7 points pooled—but it does not reorder the triggers except at $n = 20$, where all four fall within a point of each other (12.1–12.9%) and the comparison carries no information.

4.2 Why the degeneracy trigger collapses

Figure 2 explains the collapse and vindicates the concern of Luersen and Le Riche (2004) about their tolerances. Under the combined trigger the share of restarts fired by the degeneracy test rather than the size test rises from 60.7% at $n = 2$ to 84.9%, 93.2% and finally 100.0% at $n = 20$: at that dimension GBNM *is* its degeneracy test, the size half never firing at all. Meanwhile the cost diverges. The degeneracy trigger fires 463 times per run at $n = 20$ against 7.3 for a size test—a 63-fold difference, in the direction of the worse method. Each restart discards the current simplex, so the method spends its budget rebuilding rather than descending.

This is the failure mode Luersen and Le Riche anticipated: a shrinking simplex is tagged degenerate before it is tagged small. Their two-consecutive-detections rule mitigates a problem whose magnitude, as far as we can determine, has not previously been measured, and which grows without bound in dimension.

4.3 Is the failure a tuning artifact? A calibration of ε_{s3} and ε_{s4}

The indictment in Section 4.2 is only worth making if the failure belongs to the criterion rather than to the tolerances we happened to run it at—the more so because its authors call those tolerances hard to tune. We therefore swept both, log-spaced from fully disabled to maximally aggressive, holding the initializer at the regular simplex and comparing against a plain size test

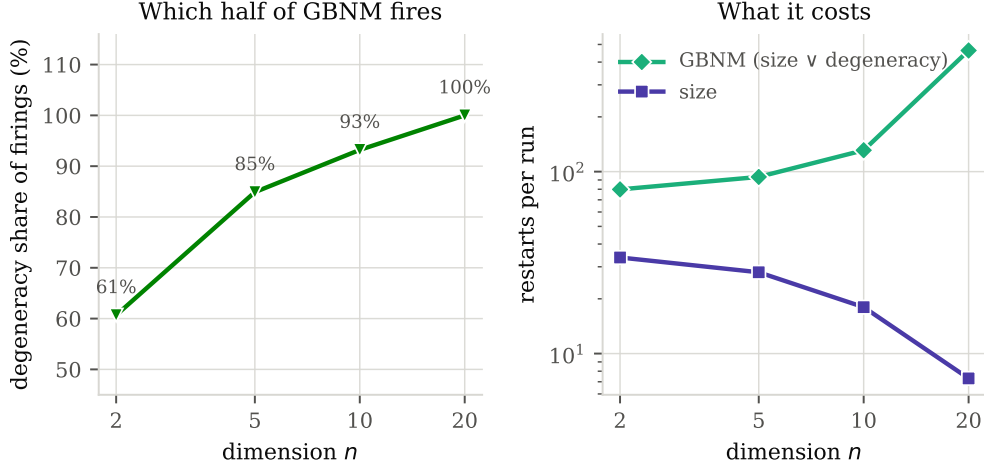


Figure 2: Left: the share of GBNM restarts fired by the degeneracy half of Eq. (2) rather than the size half. Right: what that costs. As dimension grows GBNM becomes its degeneracy test, and that test thrashes.

Table 2: Tolerance calibration. Win rate against a size test (% of problems where the degeneracy trigger ends lower; 50% = parity) and restarts per run. ε_{s3} is held at 10^{-12} : it is inert in 11 of 12 cells (see text), so the table is indexed by ε_{s4} alone.

ε_{s4}	$n = 5$		$n = 10$		$n = 20$	
	win	restarts	win	restarts	win	restarts
0 (disabled)	15.0%	0	20.8%	0	15.8%	0
10^{-40}	33.3%	58.1	39.2%	11.9	54.2%	13.0
10^{-20}	30.8%	60.7	44.2%	18.3	42.5%	44.8
10^{-12}	31.7%	66.4	37.5%	39.4	27.5%	111.1
10^{-6} (default)	27.5%	86.3	40.8%	122.1	20.0%	442.5
10^{-4}	28.3%	114.7	35.0%	238.4	10.0%	1669.2
10^{-2}	28.3%	252.9	20.8%	1148.5	10.0%	1904.0

in the same harness. The response is the fraction of problems on which the degeneracy arm ends lower than the size arm; 50% is parity. Table 2 reports it, with the resulting restart rate alongside.

Three things follow.

ε_{s3} does almost nothing. Across four orders of magnitude, 10^{-12} to 10^{-2} , the edge-ratio half of Eq. (2) leaves both the win rate and the restart rate unchanged in 11 of the 12 (dimension, ε_{s4}) combinations tested, to the digits shown. The single exception is $n = 5$ at $\varepsilon_{s4} = 10^{-12}$, where loosening ε_{s3} to 10^{-2} moves the win rate from 31.7% to 27.5%; at $n = 10$ and $n = 20$ it is inert everywhere. The Hadamard half does essentially all the work, so in practice GBNM’s two tolerances are one knob—worth knowing for anyone trying to tune it. Table 2 is therefore indexed by ε_{s4} alone, at $\varepsilon_{s3} = 10^{-12}$.

The default is badly mis-set at high dimension. Moving ε_{s4} from the default 10^{-6} down to 10^{-40} raises the win rate at $n = 20$ from 20.0% to 54.2%—34 points—and cuts the restart rate from 442 per run to 13. The trigger is not inherently a thrashing mechanism; it is a correctly-shaped criterion set thirty orders of magnitude too loose for a degenerating simplex, whose Hadamard ratio routinely reaches 10^{-40} . This is the effect Luersen and Le Riche describe

qualitatively.

But no setting makes it worth having. Testing all 21 cells against parity with a two-sided binomial test and a Bonferroni correction ($\alpha = 0.05/21$), *none* is significantly better than a plain size test at any dimension. The single above-parity cell— $n = 20$ at $\varepsilon_{s4} = 10^{-40}$, 54.2%—has $p = 0.41$ and is indistinguishable from a coin flip. Eleven of the 21 cells are significantly *worse*. Disabling the criterion outright is worse still (15–21%), so restarting does useful work; it is the degeneracy criterion specifically that adds nothing a size test does not already provide.

The conclusion of Section 4.2 therefore survives, but its scope changes. “Actively harmful” is a statement about the published default, not about the criterion: correctly tuned, the trigger is merely redundant. Both readings argue for the same practical action, which is to drop Eq. (2) and keep the size test.

One caveat bounds this section. The sweep holds the initializer fixed at the regular simplex and uses a single algorithm seed, so the calibrated optimum is strictly “best for that initializer”. Since the finding is a null—no setting helps—an initializer-specific optimum weakens it less than it would a positive result.

4.4 The two natural metrics disagree at $n = 20$

Because f_L is the best value reached by any arm, an arm that wins *outright* on a problem is automatically credited with solving it at every τ . This is the Moré and Wild (2009) convention and is standard, but it rewards occasional large wins. Kelley’s arm attains f_L on 27% of $n = 20$ problems, more than any other arm, which is why it leads Table 1 there.

An f_L -free measure tells a different story—in fact it reverses the sign. Comparing Kelley against no-restart problem by problem and asking simply which ends lower, Kelley wins 47.5% of the time at $n = 20$, i.e. slightly less than half, while the solved-fraction has it ahead 26.4% to 11.4%. Both numbers are correct: Kelley reaches the best value of any arm on 27% of problems and is mediocre on the rest.

That pairwise view also exposes an interaction the pooled figures hide. Kelley’s advantage depends strongly on what it is rescuing: against no-restart at $n = 20$ it wins 82.5% of problems started from the `fminsearch` simplex, but only 40.8% from the large regular simplex. An oriented restart is most valuable exactly where initialization was worst, which is the substitution result of Section 4.5 reappearing at the level of a single method. Ranking the ten arms by final objective within each problem and averaging, the $n = 20$ order is **size** and **flat** at 4.28, **size/oriented** at 4.92, **progress** at 5.18, the degeneracy arms at 6.11, and **none** last at 6.95. So Kelley wins big but inconsistently, while the trivial triggers win small but reliably; both readings are correct and they answer different questions. We report both rather than choosing. The disagreement is confined to $n = 20$; at $n \leq 10$ the two measures agree on the ordering.

Verification against absolute targets. The principled resolution is not to arbitrate between the two but to remove the dependence on the compared set. Although this COCO build does not expose f_{opt} through its public interface, the optimal parameter can be recovered per problem and evaluated, giving f_{opt} exactly; we did so for all 480 problems, and confirmed that no run in the study ever beat it. That permits the standard BBOB precision target $\Delta f = f - f_{\text{opt}} \leq 10^{-8}$, which is absolute.

Both headline trends survive. The share of discordant problems favouring Kelley over no-restart still rises with dimension (Cochran–Armitage $z = +3.46$, $p = 5.4 \times 10^{-4}$), and for the degeneracy trigger still falls ($z = -6.08$, $p = 1.2 \times 10^{-9}$); the degeneracy trigger’s absolute solve rate at $n = 20$ is 0.1% against 3.7% for never restarting, a starker gap than the relative measure shows. What weakens is the *point* comparisons: at $n = 20$ scarcely any arm reaches 10^{-8} , so there are few discordant pairs and little power, and neither Kelley’s advantage nor the degeneracy

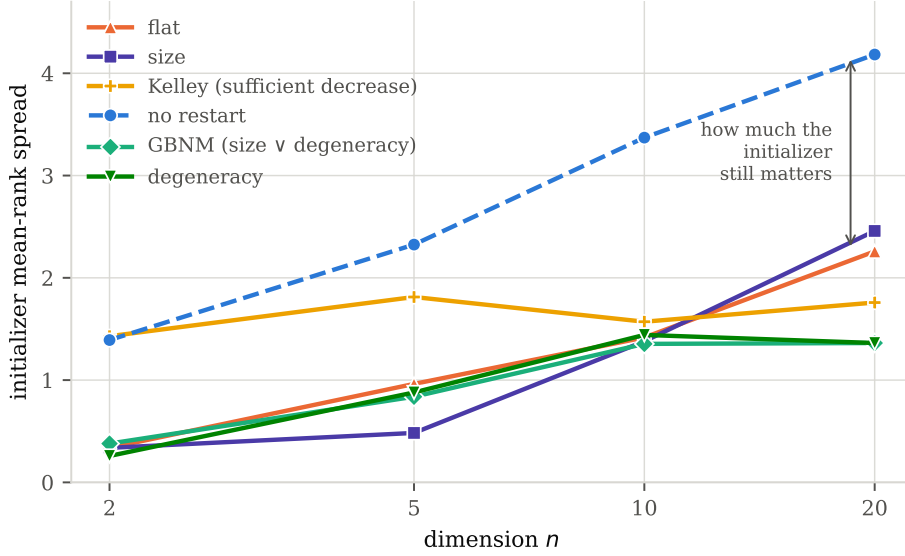


Figure 3: Spread in initializer mean rank against dimension. Under a single descent the choice of initializer matters more and more; any working trigger holds it far lower, though decreasingly so.

trigger’s deficit is individually significant there. This is the same lesson as Section 4.1—the dimensional claims are trend claims and should be read as such—and it now holds under two independent definitions of “solved”.

4.5 Restarts substitute for initialization

Table 3 ranks the seven initializers within each problem. The main effect is overwhelming in every row (Friedman, $p = 2 \times 10^{-124}$ for **none**), so what follows is about magnitude, not power.

Enabling any trigger collapses the spread—from 2.68 pooled under a single descent to 0.89–1.24—and the clearest case is **pfeffer**, worst of the seven at 5.56 under a single descent, reproducing the central claim of Wessing (2019) on his own example, and mid-pack at 3.88–4.25 under any trigger. **progress** is intermediate at 1.44: a local re-seed absorbs less of the initialization effect than a global one, as one would expect.

The substitution erodes with dimension (Figure 3). Under a single descent the spread grows monotonically, $1.39 \rightarrow 2.32 \rightarrow 3.37 \rightarrow 4.18$; under a working trigger it also grows, $0.34 \rightarrow 0.48 \rightarrow 1.38 \rightarrow 2.26$. Restarts still absorb most of the effect at every dimension, but the absolute residual at $n = 20$ is larger than the entire effect was at $n = 2$. One caution on reading Figure 3: the degeneracy arms show a *low* spread at $n = 20$ (1.36) not because they absorb the initialization effect but because they perform uniformly badly, which compresses ranks. Spread is only interpretable for triggers that work.

Across every trigger and dimension the best initializer is the large regular simplex. We do not rank the two random initializers against each other: BBOB’s box is $[-5, 5]^n$ for every problem, so at a fixed algorithm seed **uniform** and **gaussian** draw the *same* initial simplex on every problem of a given dimension. Their rows therefore rest on one realization of the initializer, and only the instance-to-instance shift of the optimum randomizes the comparison. The deterministic initializers—which carry both claims above—are unaffected, and the trigger comparisons pool over all seven initializers, so a lucky or unlucky draw enters every trigger arm equally and cancels.

Table 3: Initializer effects. *Top*: mean rank (1 = best of 7), pooled over dimension—pooled because the *ordering* of initializers is dimension-stable. *Bottom*: the spread max – min, which is not, so it is broken out by dimension.

trigger	pfeffer	sp. 100%	sp. 10%	sp. 2%	uniform	gaussian	p. basis
none	5.56	2.88	3.69	4.54	3.49	4.57	3.26
progress	4.63	3.19	3.86	4.37	3.84	4.57	3.55
flat	4.25	3.34	3.91	4.02	4.03	4.58	3.87
size	4.17	3.49	3.85	4.12	3.97	4.58	3.82
conditioning	3.89	3.58	3.84	3.90	4.17	4.54	4.07
gbnm	3.88	3.65	3.86	3.88	4.17	4.54	4.01

spread by dimension	$n=2$	$n=5$	$n=10$	$n=20$
none	1.39	2.32	3.37	4.18
progress	1.43	1.81	1.57	1.76
flat	0.34	0.96	1.42	2.26
size	0.34	0.48	1.38	2.46
conditioning	0.26	0.88	1.44	1.36
gbnm	0.38	0.84	1.35	1.36

4.6 Conditioning does not mediate the effect

The intuitive account is that initialization sets simplex conditioning, contraction destroys it, and restarts reset it. The data do not support it. The Spearman correlation between a run’s median Hadamard ratio and its final objective, computed within (function, dimension, trigger) and pooled, is -0.063 , -0.016 , -0.031 and -0.101 at $n = 2, 5, 10, 20$, with 61%, 58%, 56% and 67% of cells negative. We report this because it is the natural hypothesis, and because it is the most likely reason the degeneracy trigger underperforms: a criterion measuring a quantity only weakly coupled to progress cannot identify a good time to restart. The measurement is correlational and within-cell, and does not establish or exclude a causal role.

4.7 The dimensional story is not a budget artifact

Everything above runs at a budget of $2000n$, and at $n = 20$ the no-restart arm solves only 11.4% of problems, so the dimension might be budget-starved rather than hard—which would make the trends a statement about the budget instead of about n . We re-ran six arms at $n \in \{10, 20\}$ under both $2000n$ and $4000n$, holding the initializer at the regular simplex.

Quadrupling the effective budget changes nothing material. The mean rank of the six arms within each problem gives an *identical* ordering at both budgets and both dimensions (Kendall $\tau = 1.000$), no arm shifting by more than 0.21 rank; win rates against no-restart move by at most a few points and never change sign. The one systematic effect is that a larger budget mildly favours the cheaper triggers—**size** improves from 75.8% to 89.2% against no-restart at $n = 20$ —which sharpens rather than softens the paper’s recommendation.

This run also supplies a comparison the tolerance sweep of Section 4.3 could not make, because that sweep contained no no-restart arm. At its best tolerance the degeneracy trigger beats no-restart on 84–96% of problems at both dimensions and both budgets. So the indictment in Section 4.2 is emphatically a statement about the published default: retuned, the criterion is a perfectly serviceable restart trigger that happens to be no better than testing whether the simplex is small (39.2%–54.2% against **size**, indistinguishable from parity at three of four dimension–budget combinations). At the default it is far worse than either (20.0–24.2% against **size**).

4.8 A caution about unshifted test functions

On the five classical analytic functions, over $n \in \{2, 5, 10\}$, the same experiment reverses Section 4.5: the spread is 3.80 under a single descent and 3.60–3.82 under four of five triggers—no substitution at all.

The explanation is that sphere, Rosenbrock, Rastrigin, Ackley and Griewank all place their optimum at or near the centre of the conventional box, so a structured simplex built about that centre begins at the answer. Restarts cannot improve on an initializer that has already won, and the interaction is suppressed by construction. These functions remain common in the Nelder–Mead variant literature; for questions about initialization they are not merely weak but actively misleading, and a shifted, rotated suite is required.

5 Conclusions

On (Q1): restarts substitute for initialization, collapsing the spread across seven initializers from 2.68 to 0.89–1.24 and rendering the `fminsearch` default unremarkable. The substitution erodes with dimension, so Wessing’s advice is strongest exactly where it is hardest to avoid needing it. This qualifies Wessing (2019) rather than contradicting him, and reconciles his result with the expensive-objective regime of Takenaga et al. (2023), where restarts are unaffordable and initialization is consequently decisive.

On (Q2): the honest answer is that it depends on n , and neither published detector is a safe default across the range. For $n \leq 10$ a simplex-flatness or simplex-size test dominates everything else tested. At $n = 20$ the trivial triggers remain the most consistent, but Kelley’s detector attains the best value on more problems than any other arm, and the degeneracy test of Luersen and Le Riche (2004) has become actively harmful—worse than never restarting, firing 463 times per run, and by then accounting for 100% of GBNM’s restarts. A practitioner using GBNM at moderate dimension should disable Eq. (2) and keep the size test—and Section 4.3 shows this is not a tuning failure they can repair: across 40 orders of magnitude of ε_{s4} , no setting is significantly better than the size test at any dimension, and ε_{s3} is very nearly inert. If a construction that shrinks the simplex is used, it must be paired with a trigger that does not fire on smallness—which is exactly what Kelley (1999) does.

Three limitations bound these claims. The dimensional trends rest on four points with one instance seed per problem, and the $n = 20$ conclusions are the least well supported—though Section 4.7 shows they are at least not an artifact of the evaluation budget. Restart construction has two levels; GBNM’s own probabilistic re-seeding from a kernel density over past search points is not implemented, and the factorial test of O’Neill (1971) is not among the triggers compared. The f_L convention that complicates Section 4.4 is discharged rather than merely disclosed: every headline claim was re-verified against absolute targets recovered from f_{opt} , and the trends hold under both.

Reproducibility. Implementation, raw per-run records for all 43 050 runs, the recovered f_{opt} values, and the analysis scripts are available at <https://github.com/Mollinetti/nelder-mead-init-restart>. Results were produced with Python 3.14.6, NumPy 2.x, SciPy 1.x and `coco-experiment` 2.8.2 on macOS/arm64; every algorithm run uses seed 101, and the per-run records carry the initializer, trigger, construction, tolerances and budget needed to reproduce any single cell.

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